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by Phil

General metrics

17,779	2,591	147	10 min 21 sec	19 min 55 sec
characters	words	sentences	reading time	speaking time

Score

78

This text scores better than 78% of all texts checked by Grammarly

Writing Issues

149	46	103
Issues left	Critical	Advanced

Writing Issues

5 **Clarity**

5 Wordy sentences

46 **Correctness**

22 Mixed dialects of english

4 Faulty subject-verb agreement

1 Closing punctuation

1 Conjunction use

3 Determiner use (a/an/the/this, etc.)

5	Comma misuse within clauses
2	Incorrect noun number
2	Wrong or missing prepositions
2	Misspelled words
2	Improper formatting
2	Confused words

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

26%

unique words

Rare Words

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

45%

rare words

Word Length

Measures average word length

5.6

characters per word

Sentence Length

Measures average sentence length

17.6

words per sentence

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Chapter 3 Research Design and Methodology

3.1 Introduction

In Chapter 1, readers will be introduced to the essential building blocks of the research inquiry and given an overview of what the dissertation will cover. The research design is crucial as it outlines how various study components will be brought together in a logical and cohesive way[1] to effectively address the research problem. It acts as a roadmap for data collection, measurement, and analysis (Sileyew, 2019). The chapter aims to provide a detailed overview of the DSRM research methodology and design that will be employed in this study while also explaining the reasoning behind its selection. In this chapter, we will discuss the activities and guidelines of the DSRM that will be used to answer the research questions of this study. The chapter includes the motivation for selecting DSRM (Section 3.2), the type of artifact in DSR and the category it falls under for this study (Section 3.3), the DSRM process that will be adopted for the research (Section 3.4), and the research methods that will be used in this study (Section 3.5). Section 3.6 will outline the Application of DSRM and Dissertation Structure.

3.2 Motivation for Selecting DSRM for this study.

Design science is a scientific field that studies the development and use of artefacts[2] to solve practical problems (Simon, 1996). Design Science Research (DSR) is an approach to problem-solving that aims to advance human knowledge by developing inventive solutions. It centers around creating artifacts that have the potential to improve technology and scientific

knowledge bases. DSR not only generates newly designed artifacts but also yields invaluable design knowledge (DK) that grants a deeper comprehension of how these artifacts can either enhance or disrupt relevant contexts. Essentially, DSR strives to ameliorate existing knowledge by generating innovative solutions. (Hevner, March, Park, & Ram 2004 By applying design theories, Design Science Research (DSR) offers a holistic perspective on the impact of novel solutions. The main objective of a DSR research endeavor is to enhance human and organisational[3] abilities through the development of distinctive artifacts in the shape of models, constructs, methods, and instantiations. (Hevner et al. 2004, Gregor & Hevner 2013). DSR intends to develop knowledge about how things can be designed or arranged (i.e., design knowledge or DK) to achieve a specific set of objectives. This design knowledge is typically generated by human agency.

DSR assists in bridging the gap between the physical artefact[4] and the accompanying theory. DSR aims to produce and communicate knowledge that is relevant to a global interest, and not only to a secluded group of individuals, which is often the focus on design-intensive projects (Johannesson & Perjons 2012). DSR enables the exploration and building of artefacts[5] in a series of iterations and assists with answering complex research questions. Research questions need to be developed on a theoretical basis, which guide[6] the experimental design of field experiments and evaluations (Vaishnavi & Kuechler 2015). When DSR is applied to IS research, a body of knowledge is established about artificial objects and phenomena to meet specific needs and goals. The main research question to be answered in this study is how can patient data management processes at a clinic be improved with the implementation of RPA?[7] The DSR research design methodology will aid in answering this question by providing a platform and a research method that

allows for the development of an artefact[8] to be able to aid in answering the research question.

3.3 Artefact Types

An artefact[9] is a thing that researchers have made with the intention of solving[10] a problem that already exists in the context. Designing, producing, and analysing[1] artefacts[12] are common research activities in IT and computer science research, according to March and Smith (1995). Because there are different kinds of problems and different kinds of contextual environments in which these problems exist, there will always be a variety of artefacts[13] presented to handle the issue. According to March and Smith (1995), artefacts are research results, but Hevner and[14] (Chatterjee, 2010) define[15] them as the ultimate objective of design science research initiatives. Research results, or artefacts,[16] fall into several categories, including instantiations, methodologies, models, constructs, and improved design theories. and structures.

- Methods are a set of steps explaining how to achieve something, which includes algorithms and practices. They encompass the methodologies, tools, and processes employed during the research process. Methods ensure that the artifact is developed systematically and rigorously. In DSR, methods help guide the creation and assessment of the artifact and contribute to the research's credibility and validity (Johannesson and Perjons, 2012).
- Models are propositions that describe a set of constructs to solve a problem, while constructs refer to concepts, syntax, or language used in a specified context to describe a problem and find a solution. They are built up from constructs that are related to each other, and they are

presentation[17] of possible solutions to practical problems (Johannesson & Perjons, 2014).

- Constructs: Concepts, syntax or[18] language (vocabulary and symbols) used in a specified context to describe a problem and find a solution are examples of constructs. They are essential for specifying what is being investigated or manipulated during the research process. In DSR, constructs help researchers articulate the essential components of their research and the factors they consider important when designing and evaluating their artifact.[19]
- Instantiations refer to the realisation[20] of an artefact[21] in IT, such as system implementation or prototype development. Instantiations can be a working system or solution that is applied in the contextual environment to solve existing problems. Instantiations show that constructs, models and[22] methods can be implemented in a working system (Vom Brocke, 2017). These instantiations, in the form of the RPA processes being developed, directly address the real-world problem of automating patient data management in a clinical setting. They will serve as the tangible and implementable results of the research, showcasing the innovative solution designed and demonstrating the practical impact of the work within the healthcare domain. Instantiation is about putting the designed solution into practice to see how it performs and whether it effectively addresses the problem (Mijač, M. 2019). Instantiations can also appear as purely or predominantly technical artifacts, which means they require no or minimum of[23] interaction with humans (e.g. software components embedded into larger, possibly socio-technical artefact[24]) Gregor and Jones (2012).

DSR Cycles

The relevance cycle, the design cycle, and the rigor cycle are the three distinct research cycles that make up[25] design research (Hevner et al., 2010). The research challenge, or the necessity for the research, as well as the research environment, are described in the relevance cycle. The rigour[26] cycle provides a foundation for rigorous design research by drawing on existing knowledge bases such as theories, techniques, design products, design processes, artifacts, experiments, and expertise. The research activities and actions are made up of the design cycle.

1.Relevance[27] Cycle

This cycle starts with the discovery and comprehension of a pertinent, practical problem or opportunity in a particular domain. The context, significance, and influence of the problem on stakeholders are all things that researchers hope to fully comprehend. The aims and goals of the research are set during this phase. By creating a brand-new item or solution, they define what they hope to accomplish. The artifact is created and designed by researchers to answer the recognized issue or opportunity. The creation of the artifact aims to make it usable and relevant to the context of the real world. Researchers show the artifact's functioning and relevance by demonstrating how it can successfully handle the defined problem once it has been created (Hevner & Chatterjee 2010).

2.Design[28] Cycle:

Analysis and Design: During this phase, researchers examine the artifact's architecture and design. They produce conceptual models, design guidelines, and development strategies for the artifact. Based on their design parameters, researchers construct prototypes or put the item into use. Creating tangible things from design concepts is the goal of this phase. The artifact is assessed to see if it satisfies the design objectives and can solve the issue successfully.

Researchers evaluate its usability and functionality. Reflecting on the results of the design and assessment phases, researchers determine what went well and what may be improved in the design of the artefact[29] Hevner. (2007).

3. Rigor Cycle:

Assessment of Artifact's Rigor: In this cycle, researchers focus on the scientific rigor of their work. They evaluate the quality and robustness of the artifact in terms of its design, development, and evaluation. Researchers assess how well the artifact can be utilized in practice and whether it provides value in addressing the problem. This involves considering its usability, efficiency, and effectiveness. Researchers reflect on the artifact's scientific rigor and practical utility, making any necessary adjustments or refinements to enhance both aspects. The findings and insights gained from the rigor cycle are communicated through academic publications, presentations, and knowledge dissemination channels. This step contributes to the broader academic community and informs practitioners about the artifact's value (Peffers et al., 2007).

The three distinct research cycles in Design Science Research (DSR) highlight the importance of relevance, design, and rigor in creating and evaluating research artefacts[30]. By iteratively going through these cycles, researchers can develop innovative solutions that not only solve real-world problems but also adhere to rigorous research standards. This ultimately contributes to both academia and practical applications. Therefore, these three distinct research cycles will be relevant for the research.

3.4 DSRM Process

To achieve the desired outcome, the DSR process comprises six critical steps: problem identification and motivation, objective definition for solution, design and development, demonstration, evaluation, and communication.

Additionally, there are four potential entry points. problem-centered initiation, objective-centered solution, design and development-centered initiation, and client/context initiation. A brief description of each DSR activity follows.

Figure 5 DSR Methodology Process Mode(Preffers et aal., 2007.[31]

Problem identification and motivation. Effective problem identification and motivation are crucial in research. It entails defining the appropriate research problem and demonstrating the significance of the chosen solution. By justifying the value of the solution, the audience and researcher are motivated to pursue it. Additionally, it enables the audience to comprehend the problem and appreciate the researcher's understanding of it. (Peffers et al., 2007). Therefore, highlighting the problem's current state and the solution's importance is imperative.

Define the objective for a solution. Careful analysis of the problem at hand and consideration of feasibility are necessary to define solution objectives. Objectives may be quantitative or qualitative in nature.[32] Quantitative objectives outline how a new solution would improve upon existing ones, while qualitative objectives address previously unaddressed issues. It is crucial to logically derive objectives from the problem definition (Peffers et al., 2007).

Design and development. In the process of design and development, an artifact is created. The artifact can be any designed object that embeds a research contribution in its design. This activity involves defining the desired functionality and architecture of the artifact,[33] and then creating the actual object. (Peffers et al., 2007).

Demonstration. This activity demonstrates how to use the artifact to solve the problem through experimentation, simulation, case study, proof, or other suitable activity. Demonstrating the artefact[34] should allow participants to interact with the created prototype or system and will provide valuable

feedback on any aspects that are unclear or require attention. The feedback is used to modify the prototype's design and functionality before advanced evaluations are conducted (Peffers et al., 2007).

Evaluation[35] process involves assessing how well the created artifact supports the solution to the problem. This step requires comparing the objectives of the solution with actual results obtained by using the artefact[36] in its intended setting. The type of evaluation required may vary depending on the nature of the problem and the artifact used. Once the evaluation is complete, researchers can either go back to step three to improve the artifact's effectiveness or proceed to communication and leave further improvement to future projects (Peffers et al., 2007).

Communication. Effective communication with relevant stakeholders employs appropriate forms depending on research goals and audience, including practitioners. Communication typically takes the form of reporting on the results and findings of the study, and may also be used for research publications, which can be submitted to journals and conferences (Peffers et al., 2007).

3.5 Research Strategies

Research strategies refer to the systematic plans and approaches[37] that researchers use to conduct investigations, gather data, and analyse[38] information in order to[39] answer specific research questions or achieve research objectives. The choice of research strategy depends on the nature of the research, the research questions, available resources, and the desired outcomes (Wedawatta G., 2011). This section looks at research strategies adopted for the research.

3.5.1 Literature studies

Literature studies are an informative way to gain an understanding of a particular topic in a research field. They provide a comprehensive overview of the various sub-components of a given field, which can be useful for combining sub-components from diverse fields to explain interdisciplinary concepts. In this study, literature studies will be utilised[40] to examine existing research and solutions for related issues (Saunders et al. 2009). However, it is important for researchers to not solely rely on literature to comprehend a problem. Engaging with participants and stakeholders of relevant practices is also crucial. The study will gather information on RPA, AI, hospital systems, healthcare, hospitals, clinics and[41] information systems from various sources such as white papers, journals, articles, and newspapers.

3.5.2 Case study

Case study research involves an extensive analysis of a single case, such as an organisation,[42] or a small number of cases. In this type of research, information is collected from various sources and through different types of data, such as observations, surveys, interviews, and analysis of documents. The data can be qualitative, quantitative, or a combination of both. Case study research enables a comprehensive and multifaceted examination of the issue or problem at hand. For the study, a specific clinic will be selected for observation, interviews, analysis, testing and[43] implementing[44] RPA[45] processes for managing patient data (Johannesson & Persons, 2012).

3.5.3 Extant Systems Analysis

Extant System analysis involves systematic[46] and comprehensive investigative process employed to assess and comprehend the operational characteristics, intricacies, and performance metrics of existing systems, processes, or

structures within a defined domain (Satzinger, 2015). The fundamental purpose of extant systems analysis is to gain a holistic understanding of the status quo, highlighting both strengths and weaknesses, in order to[47] inform subsequent decisions, enhancements, or transformations (Tilley, 2019). Through the diligent application of extant systems analysis, researchers aim to extract valuable insights, which can serve as a foundational basis for proposing improvements, optimisations,[48] or strategic changes within the studied system. (Lim, 2006) explains that an extant systems analysis is performed to collect background information to support the development of a new, or improved, system from users, or other sources. For the research an extant system analysis will be carried out on the case study to analyse[49] the functionality of the existing patient data management systems.

3.5.4 Software Evaluation

Software evaluation is a research strategy employed within the field of research methodology, characterized by a systematic and comprehensive examination of software applications or systems to assess their quality, performance, functionality, and suitability for specific purposes (Jadhav, 2009). This strategy involves the structured collection and analysis of data, often guided by predefined criteria and objectives, to make informed judgments about software products or solutions. Software evaluation serves as a critical component of research efforts, informing decisions related to software selection, usability, security, and overall effectiveness in meeting project or orginsational[50] requirements. This research strategy will be carried out to investigate RPA software that are[51] suitable to automate patient data management processes (Gediga, 2002).

1.	logically and cohesively	Wordy sentences	Clarity
2.	~~artefacts~~ → artifacts	Mixed dialects of English	Correctness
3.	~~organisational~~ → organizational	Mixed dialects of English	Correctness
4.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
5.	~~artefacts~~ → artifacts	Mixed dialects of English	Correctness
6.	~~guide~~ → guides	Faulty subject-verb agreement	Correctness
7.	~~RPA?~~ → RPA.	Closing punctuation	Correctness
8.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
9.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
10.	to solve, intending to solve	Wordy sentences	Clarity
11.	~~analysing~~ → analyzing	Mixed dialects of English	Correctness
12.	~~artefacts~~ → artifacts	Mixed dialects of English	Correctness
13.	~~artefacts~~ → artifacts	Mixed dialects of English	Correctness
14.	~~and~~	Conjunction use	Correctness
15.	~~define~~ → defines	Faulty subject-verb agreement	Correctness
16.	~~artefacts~~ → artifacts	Mixed dialects of English	Correctness
17.	the presentation, or a presentation	Determiner use (a/an/the/this, etc.)	Correctness
18.	, or	Comma misuse within clauses	Correctness
19.	~~artifact~~ → artifacts	Incorrect noun number	Correctness
20.	~~realisation~~ → realization	Mixed dialects of English	Correctness

21.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
22.	, and	Comma misuse within clauses	Correctness
23.	~~of~~	Wrong or missing prepositions	Correctness
24.	~~artefact~~ → artefacts	Incorrect noun number	Correctness
25.	~~make up~~ → makeup	Misspelled words	Correctness
26.	~~rigour~~ → rigor	Mixed dialects of English	Correctness
27.	. Relevance	Improper formatting	Correctness
28.	. Design	Improper formatting	Correctness
29.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
30.	~~artefacts~~ → artifacts	Mixed dialects of English	Correctness
31.	~~aal~~ → al	Confused words	Correctness
32.	~~in nature~~	Wordy sentences	Clarity
33.	artifact,	Comma misuse within clauses	Correctness
34.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
35.	The evaluation	Determiner use (a/an/the/this, etc.)	Correctness
36.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
37.	~~approaches~~ → approach	Faulty subject-verb agreement	Correctness
38.	~~analyse~~ → analyze	Mixed dialects of English	Correctness
39.	~~in order to~~ → to	Wordy sentences	Clarity
40.	~~utilised~~ → utilized	Mixed dialects of English	Correctness
41.	, and	Comma misuse within clauses	Correctness

42.	~~organisation~~ → organization	Mixed dialects of English	Correctness
43.	, and	Comma misuse within clauses	Correctness
44.	~~implementing~~ → implementation	Confused words	Correctness
45.	of RPA	Wrong or missing prepositions	Correctness
46.	a systematic	Determiner use (a/an/the/this, etc.)	Correctness
47.	~~in order to~~ → to	Wordy sentences	Clarity
48.	~~optimisations~~ → optimizations	Mixed dialects of English	Correctness
49.	~~analyse~~ → analyze	Mixed dialects of English	Correctness
50.	~~orginsational~~ → organisational, organizational	Misspelled words	Correctness
51.	~~are~~ → is	Faulty subject-verb agreement	Correctness